

Is it possible to determine the critical phase transition temperature in a 2D infinite Ising model using a finite-size approximation simulation?

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In this article, we attempted to determine the critical temperature for phase transitions in the infinite 2D Ising model. We used a Monte Carlo approach with the Metropolis algorithm. We then use the value of heat capacity of the model to determine the value of the critical temperature as at the critical temperature, the heat capacity function exhibits a singularity where it goes to infinity[1]. We first determined upper and lower bound for the critical temperature using small simulations where we look at the heat capacity and determined whether the chosen value of temperature corresponds to a heat capacity above or below that of the critical temperature. We then ran a comprehensive multi-simulation over the predetermined temperature interval to find the critical temperature value. We obtained a value of $T_c = 2.2834 \pm 0.0009$, with the theoretical value given by $T_c = \frac{2}{\ln(1+\sqrt{2})} \approx 2.269$.

We encountered a limitation of the finite-size approximation of the infinite lattice, which should be addressed in further research.

I. INTRODUCTION

A. The Ising model

The Ising model named after the physicist Erns Ising is a model in statistical mechanics. The model consists of discrete magnetic spin dipole moments that take values ± 1 ("up" and "down" respectively) arranged on a lattice. For the purposes of our simulation we used a square lattice. The spins interact with only a subset of all the spins that are "close", we call this subset the neighbouring spins. It is known that this model exhibits phase transitions that can be associated with the gas-liquid transition or spontaneous magnetisation. [2]

B. Aim of study

In this article we look at determining at what temperature does the Ising model undergo a phase transition. It is known from theory that below a certain critical temperature and after sufficient amounts of time, the model stabilises into completely up or down state. Conversely, above this critical, on average, the model remains a random mixture of up and down spins. [2] We will try to determine this value by simulating the Ising model for different temperatures, differentiate between ordered and random results, and determine the critical value at which the phase transition happens.

II. THEORY

In the Ising model, the lattice spin elements take the values $s_i = \pm 1$. The interaction energy between two spins is given by

$$E_{\text{int}} = -J s_i s_j, \quad (1)$$

where $J > 0$ is the coupling constant of ferromagnetic interactions and the s_i being the value of spin i .

We define the reduced interaction energy as $E_r = \frac{E_{\text{int}}}{J} = -s_i s_j$. The reduced interaction energy with the j nearest neighbours is then

$$E_r = \sum_{\langle j \rangle} s_i s_j, \quad (2)$$

The energy difference between the current spin at some position and the energy after the spin has been flipped is given by

$$\Delta E = E_{\text{before}} - E_{\text{after}} = \sum_{\langle j \rangle} (-s_i) s_j - \sum_{\langle j \rangle} s_i s_j = -2 \sum_{\langle j \rangle} s_i s_j. \quad (3)$$

The total amount of energy of the whole spin lattice is given by the sum over all pairwise interaction energies. This is then denoted as

$$E_{\text{sys}} = -J \sum_{\langle i, j \rangle} s_i s_j, \quad (4)$$

where the sum over i, j counts every pair only once with $(i, j) \equiv (j, i)$. The reduced system energy is then given by

$$E_{\text{sys-red}} = \frac{E_{\text{sys}}}{J} = - \sum_{\langle i, j \rangle} s_i s_j, \quad (5)$$

We then define another dimensionless quantity, the reduced temperature as

$$T_r = \frac{k_B T}{J}, \quad (6)$$

where T is the physical temperature, k_B is the Boltzmann's constant and J the coupling constant. We then only work in terms of this reduced temperature and give no physical values of T or J .

The theoretical value of the critical reduced temperature was predicted by Lars Onsager to be

$$T_c = \frac{1}{H_c} = \frac{1}{\frac{1}{2} \ln(\cot(\frac{\pi}{8}))} = \frac{2}{\ln(1 + \sqrt{2})}, \quad (7)$$

where $H = J_r$ is the reduced coupling constant and H_c is the critical coupling constant[1].

Later, in 1952, Yang Chen-Ning showed that the function of absolute value of average magnetisation as a function of reduced temperature is given by

$$|\langle m \rangle|(T_r) = \left[\frac{1 + x^2}{(1 - x^2)^2} (1 - 6x^2 + x^4)^{\frac{1}{2}} \right]^{\frac{1}{4}}, \quad (8)$$

with $x = e^{-\frac{2}{T_r}}$ [3].

The heat capacity of the Ising model is given by

$$C = k_B \beta^2 (\langle E^2 \rangle - \langle E \rangle^2), \quad (9)$$

where $\beta = \frac{1}{k_B T}$ [2]. We can modify to equation to get an expression for the heat capacity in terms of the reduced energy E_r and reduced temperature T_r as

$$\begin{aligned} C &= k_B \beta^2 (\langle E^2 \rangle - \langle E \rangle^2), \\ &= k_B \frac{1}{k_B^2 T^2} (\langle E^2 \rangle - \langle E \rangle^2), \\ &= k_B \frac{1}{J^2 T_r^2} (\langle E^2 \rangle - \langle E \rangle^2), \\ &= k_B \frac{1}{J^2 T_r^2} \mathcal{J}^2 (\langle E_r^2 \rangle - \langle E_r \rangle^2), \\ &= k_B \frac{(\langle E_r^2 \rangle - \langle E_r \rangle^2)}{T_r^2}. \end{aligned}$$

We define the dimensionless reduced heat capacity as

$$C_r = \frac{C}{k_B} = \frac{(\langle E_r^2 \rangle - \langle E_r \rangle^2)}{T_r^2}. \quad (10)$$

We know that the heat capacity has a singularity at the critical temperature where it becomes infinite[1].

An approximation of the heat capacity around the critical temperature is given by

$$C \approx B - A \ln(\sqrt{(T - T_c)^2}). \quad (11)$$

III. METHODOLOGY

A. Metropolis algorithm

To simulate the evolution of the Ising model we use a Monte Carlo simulation using the Metropolis algorithm. In our implementation we consider the von Neumann neighbourhood of only 4 neighbouring spins in the $\pm x$ and $\pm y$ directions. We also use a periodic boundary condition such that if we denote by $n \times n$ the size of the

lattice, we have that the spin at position (n, i) is a neighbor of spin at position $(1, i)$ and in the same way also spin at i, n is neighbouring the spin at $(i, 1)$. At each iteration of the algorithm a random position is chosen in the square lattice of spins. The energy needed to flip the spin from the up to the down state or vice versa is then calculated from Eq. (3). We then have 2 cases. If $\Delta E \leq 0$, we flip the spin orientation. If $\Delta E > 0$, we accept the flip with probability $p = e^{-\frac{\Delta E}{T_r}}$. In order to do this, we sample a random variable $X \sim Unif(0, 1)$ and check $X < p$, in which case we accept the flip and leave the spin unflipped otherwise.

Theory tells us that for values of the reduced temperature below the critical temperature i.e. $T_r < T_c$ after sufficient spin-flip attempts, the system gets to an equilibrium state where all the spins are either up or down and therefore the absolute value of the average magnetisation goes to 1[2]. To calculate the average quantities $|\langle m \rangle|$ and C_r we then wait for a certain amount of spin-flip attempts before starting to calculate them, so that we measure the equilibrium of the model. The overall workflow of the simulation program is shown in Figure 1.

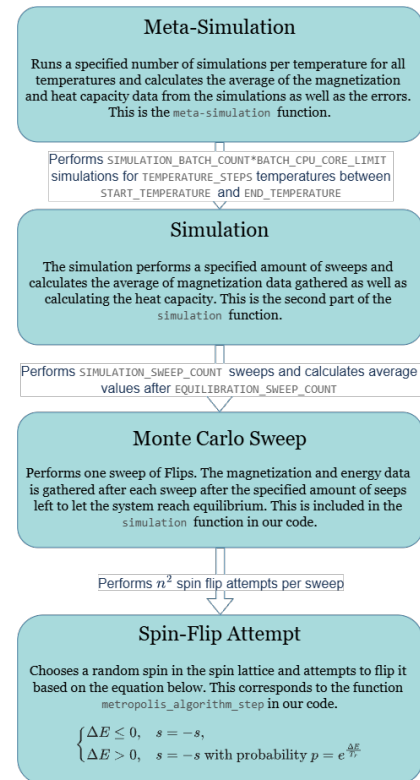


FIG. 1. Scheme of the simulation program workflow.

B. Determining the Critical Temperature

To determine the critical temperature, there are a few options. As given in the theory, the two major events

happening at this temperature is the switch of the absolute value of the average magnetization from 1 to 0, and the spike in the heat capacity. Since we are still working with some randomness, we decided that it would technically be easier to find the peak of the heat capacity. In our simulation setup, we are not finding a lot of temperatures, but we are running a lot of simulations for each temperature. This means that we do not want to use a peak finding algorithm, but rather fit a function to the data and find the maximum of that function. To determine the critical temperature, we use a fit function over the heat capacity. We use a modified version of Eq. (11), where we remove the singularity and end up with a finite peak. The formula is given by

$$C \approx B - A \ln\left(\sqrt{(T - T_c)^2 + \omega^2}\right). \quad (12)$$

This is implemented in the code as a function that takes the parameters A, B, T_c, ω and the temperature T_r and returns the value of the heat capacity at that temperature. We then use the scipy package to fit this function to the data we have collected. The parameters A, B, T_c, ω are then optimized to minimize the difference between the function and the data. The parameter T_c is then the critical temperature we are looking for. The peak of this function then gives us the critical temperature, which we can then compare to the theoretical value given in Eq. (7).

C. Determining parameters for final simulation

For the simulation that produced Figure 3 we had a number of requirements. The first requirements was for the simulation that produces the figure we use in this article to take no longer than 20 minutes to run. This meant that we had to look for simulation parameters that produce good data but for the simulation runtime to stay within the time bound.

We already knew from test simulations that the critical temperature was somewhere in the range $2 < T_c < 2.6$ as we ran simulation for $T_r = 2$ and $T_r = 2.6$ and observed the behaviour associated with reduced temperatures being lower and higher than the critical temperature respectively. This then set the range of reduced temperatures that we were going to simulate.

We then continued to choosing a size of the lattice that we want to simulate as this, we found, had impact on the number of spin-flip attempts needed for the model to reach equilibrium. We decided to use $n = 50$ as for values of $n < 20$ we found that the equilibrium was disturbed to a great degree by the non-zero value of the temperature causing fluctuations in the spins.

We then ran multiple simulations trying to ascertain the number of spin-flip attempts needed for the 50×50 lattice to reach equilibrium for the temperature range we wanted to study. We found that the configuration that took the most amounts of spin-flip attempts to reach equilibrium

was the one with the lowest temperature, namely $T_r = 2$. We decided on the number $k = 12'500'000$ for the amount of spin-flip attempts to wait before collecting values of the system energy and magnetisation for the average magnetisation and heat capacity calculations.

We then collect 10000 of these quantities to average, one every 2'500 spin-flip attempts.

Finally, we tried to run as many individual simulations for every temperature value we were interested in, while also doing the simulations for a decent enough number of different temperature values to get a good resolution in our final results. We decided to do 49 simulations for each of 62 different temperature values.

D. Uncertainties

The uncertainties in our results are calculated using the standard deviation of the mean. We calculate the standard deviation of the values we collected for the average magnetisation and heat capacity over the 49 simulations per temperature value. This gives us an estimate of the uncertainty in our results.

We also have to consider the uncertainty in our fitting function. We use the scipy package to fit our function to the data, and it provides us with an estimate of the uncertainty in the fitted parameters. We can then use this to estimate the uncertainty in our critical temperature value that is working with the uncertainties of the fit.

IV. RESULTS

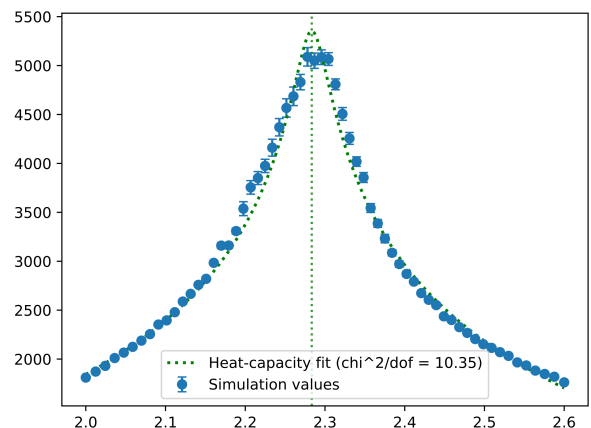


FIG. 2. Calculated heat capacity results and a fit through using function in Eq. (12) with found value of critical temperature being $T_c = 2.2834 \pm 0.0009$.

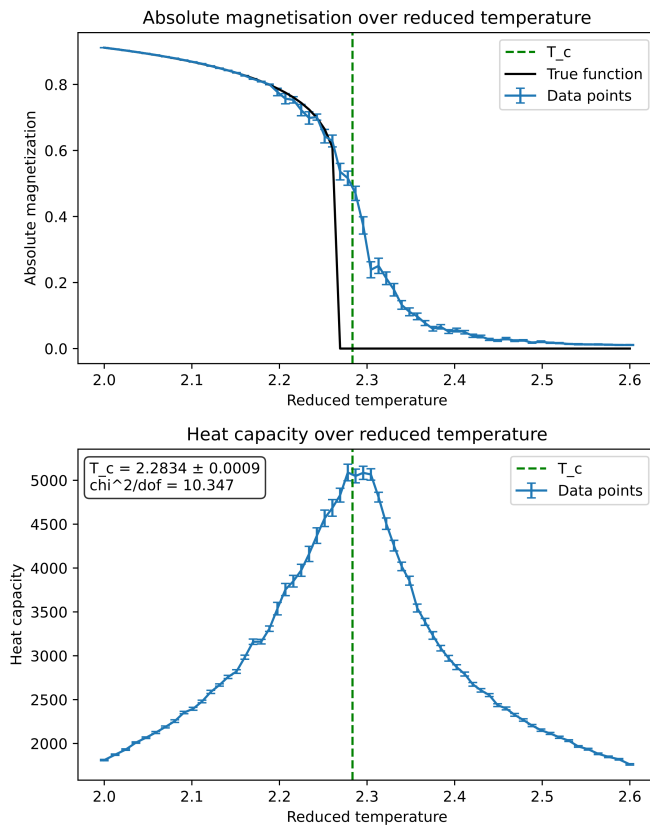


FIG. 3. Plots of average absolute magnetisation and heat capacity over the reduced temperature showing also the true function for magnetisation given by Eq. (8) and the calculated value of T_c .

The results of the heat capacity fit for parameters

$n \times n = 50 \times 50$,
 temperature range = $[2, 2.6]$,
 #temperatures = 62,
 #simulations per temperature = 49,
 #spin flip attempts per sweep = 2500,
 #sweeps for equilibration = 5000,
 #sweeps after equilibration = 10000,

are shown in Figure 2. In Figure 3 we can then see where the calculated critical temperature lies on the magnetisation and heat capacity plots.

V. DISCUSSION

We were able to determine the critical temperature of the Ising model using a Monte Carlo simulation with the Metropolis algorithm. Our results show that the critical temperature is approximately $T_c = 2.2834 \pm 0.0009$, which is in good agreement with the theoretical value of $T_c = \frac{2}{\ln(1+\sqrt{2})} \approx 2.269$. The graphical representation of the results clearly illustrates the behavior of the system as it approaches the critical temperature, with the heat capacity showing diverging behavior and the average magnetization dropping to zero. Nevertheless, the low uncertainties in combination with the high reduced chi-squared value of the fit indicate that we underestimated the uncertainties.

The difference between theoretical and experimental values of the critical temperature as well as the uncertainties might however not be due to a simulation error. We can see in the graphical representation that the heat-capacity peak is broader than the prediction for an infinite lattice. This can most likely be attributed to the finite size of the lattice used in the simulation. In a finite system, the phase transition is not as sharp as in an infinite system, leading to a rounding of the heat capacity peak and a shift in the critical temperature. This is also being discussed by M.Yu. Malsagov, I.M. Karandashev and B.V. Kryzhanovsky[4], who propose to add a correction term to the heat capacity formula, which reduces the systematic error of the finite lattice approximation. The result of this inaccuracy is that the critical temperature we determined from our simulation is slightly higher than the theoretical value as well as that the uncertainties are underestimated. This is consistent with the expectation that finite-size effects lead to a shift in the critical temperature.

VI. CONCLUSIONS

The simulation captures the qualitative behavior of the Ising model near its phase transition, but the estimated critical temperature differs from the theoretical value. Future work could improve this estimate by increasing the number of simulations, comparing alternative fitting functions and independent estimates from magnetisation, and testing different lattice sizes. The finite-size correction to the heat-capacity formula proposed by Malsagov, Karandashev, and Kryzhanovsky may also reduce systematic error; for a 50×50 lattice, they report a discrepancy below the statistical error of the Monte Carlo simulation[4].

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